TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	64.7 ± 2.72
Correlation Stability	90.4 ± 0.66
Distribution Consistency	68.1 ± 3.91
Entropy Preservation	49.1 ± 1.97
Mean Z Score Drift	0.3 ± 0.01
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	27.3 ± 6.12
Data Density	100.0 ± 0.00
Coverage Score	59.5 ± 5.50
Feature Accuracy	20.6 ± 1.50
Novelty Score	0.6 ± 0.03

TabuLLM: AI Benchmark Report

Dataset Audit: thoracic_surgery.csv

1. Dataset Portrait

Attribute	Value
Rows	470
Columns	17
Numeric features	17
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
DGN	3.096	2.883	-0.213	6.9%	OK
PRE4	3.282	3.187	-0.095	2.9%	OK
PRE5	4.569	2.794	-1.775	38.8%	Drift
PRE6	0.781	0.814	+0.034	4.3%	OK
PRE7	0.066	0.000	-0.066	100.0%	Drift
PRE8	0.145	0.011	-0.133	92.1%	Drift
PRE9	0.066	0.000	-0.066	100.0%	Drift
PRE10	0.687	0.765	+0.078	11.3%	Warn
PRE11	0.166	0.045	-0.121	72.6%	Drift
PRE14	11.736	11.739	+0.002	0.0%	OK
PRE17	0.074	0.015	-0.059	79.7%	Drift
PRE19	0.004	0.000	-0.004	100.0%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	64.7 ± 2.72
Correlation Stability (*2)	90.4 ± 0.66
Distribution Consistency (*3)	68.1 ± 3.91
Entropy Preservation (*4)	49.1 ± 1.97
Mean Z Score Drift (*5)	0.3 ± 0.01
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	27.3 ± 6.12

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	59.5 ± 5.50
Feature Accuracy (*10)	20.6 ± 1.50
Novelty Score (*11)	0.6 ± 0.03

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.851063829787234	N/A	0.6712765957446809	-0.17978723404255312	*
F1	N/A	N/A	0.1811656352530922	0.1811656352530922	*

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.847872340425532	N/A	0.5180851063829788	-0.32978723404255317	*
F1	N/A	N/A	0.18381597534108254	0.18381597534108254	*

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8553191489361703	N/A	0.526595744680851	-0.3287234042553192	*
F1	N/A	N/A	0.2167947924546989	0.2167947924546989	*

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.21063829787234045	N/A	0.22234042553191494	0.011702127659574468	
F1	0.256855620661189	N/A	0.21074672310583117	-0.046108897555357804	*

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8138297872340425	N/A	0.6904255319148935	-0.12340425531914892	*
F1	0.11972345964178996	N/A	0.16613894828550888	0.04641548864371887	

Model: DT

TabuLLM: AI Benchmark Report

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8138297872340425	N/A	0.6372340425531915	-0.17659574468085104	*
F1	0.09652882205513783	N/A	0.2187172862794211	0.12218846422428324	*

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8361702127659575	N/A	0.6893617021276596	-0.14680851063829786	*
F1	0.07414906700813283	N/A	0.20805473788418566	0.13390567087605287	*

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8510638297872342	N/A	0.5148936170212767	-0.3361702127659575	*
F1	0.011764705882352941	N/A	0.16678724479963186	0.15502253891727893	*

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.851063829787234	N/A	0.5382978723404255	-0.3127659574468085	*
F1	0.037675637056442016	N/A	0.19585776808565414	0.15818213102921214	*

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8553191489361703	N/A	0.2489361702127659	-0.6063829787234043	*
F1	N/A	N/A	0.23453654652090244	0.23453654652090244	*

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	2.0	2.88	2.16	1.0	0.0	0.0	0.0	1.0	1.0	14.0
2	3.0	3.4	1.88	0.0	0.0	0.0	0.0	0.0	0.0	12.0
3	3.0	2.76	2.08	1.0	0.0	0.0	0.0	1.0	0.0	11.0
4	3.0	3.68	3.04	0.0	0.0	0.0	0.0	0.0	0.0	11.0
5	3.0	2.44	0.96	2.0	0.0	1.0	0.0	1.0	1.0	11.0
6	3.0	2.48	1.88	1.0	0.0	0.0	0.0	1.0	0.0	11.0
7	3.0	4.36	3.28	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	2.0	3.19	2.5	1.0	0.0	0.0	0.0	1.0	0.0	11.0
9	3.0	3.16	2.64	2.0	0.0	0.0	0.0	1.0	1.0	11.0
10	3.0	2.32	2.16	1.0	0.0	0.0	0.0	1.0	0.0	11.0
11	3.0	2.56	2.32	0.0	0.0	1.0	0.0	1.0	0.0	12.0
12	3.0	4.28	4.44	1.0	0.0	0.0	0.0	0.0	0.0	12.0
13	3.0	3.0	2.36	1.0	0.0	0.0	0.0	1.0	1.0	11.0
14	2.0	3.98	3.06	2.0	0.0	0.0	0.0	1.0	1.0	14.0
15	3.0	1.96	1.4	1.0	0.0	0.0	0.0	1.0	0.0	11.0
16	3.0	4.68	4.16	1.0	0.0	0.0	0.0	1.0	0.0	12.0
17	2.0	2.21	1.88	0.0	0.0	1.0	0.0	0.0	0.0	12.0
18	2.0	2.96	1.67	0.0	0.0	0.0	0.0	0.0	0.0	12.0
19	3.0	2.6	1.68	1.0	0.0	0.0	0.0	1.0	0.0	12.0
20	3.0	2.88	2.48	0.0	0.0	0.0	0.0	0.0	0.0	11.0

TabuLLM: AI Benchmark Report

B.1 Audit Trace: Fold 1 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	3.13029465334156	2.25428743333491 5	0.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	3.42120247202026 34	2.20247106992537 76	0.0	0.0	0.0	0.0	1.0	0.0	12.0
3	2.0	4.43828892732786 1	3.00903293450562 84	1.0	0.0	0.0	0.0	1.0	0.0	13.0
4	3.0	2.52931028691486 45	1.15052861208504 4	2.0	0.0	0.0	0.0	1.0	1.0	11.0
5	3.0	2.77671158677666 96	1.97493261983499 6	0.0	0.0	0.0	0.0	1.0	0.0	12.0
6	4.0	3.21806307575661 38	2.12885095800092 83	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	2.68	2.16	0.0	0.0	0.0	0.0	0.0	0.0	12.0
8	4.0	2.32	1.76	1.0	0.0	1.0	0.0	1.0	1.0	11.0
9	2.0	2.72	2.08	0.0	0.0	0.0	0.0	0.0	0.0	12.0
10	3.0	2.32	1.76	0.0	0.0	0.0	0.0	0.0	0.0	12.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.92812195844528 04	2.19425876393649 85	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	2.0	2.85510139549705 5	2.12756571930364 66	0.0	0.0	0.0	0.0	0.0	0.0	12.0
3	3.0	4.38733576359715	3.23899635460427 4	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	2.0	3.40582222359832 2	2.46546031942617 46	1.0	0.0	0.0	0.0	0.0	1.0	11.0
5	2.0	4.77877275500800 1	3.58964027520556 4	0.0	0.0	0.0	0.0	1.0	0.0	13.0
6	3.0	2.37467278369187 97	1.97048055977991 42	0.0	0.0	0.0	0.0	0.0	0.0	12.0
7	3.0	2.84	2.24	1.0	1.0	1.0	0.0	0.0	0.0	12.0
8	3.0	3.36	2.8	1.0	0.0	0.0	0.0	1.0	1.0	12.0
9	3.0	2.44	1.96	1.0	0.0	1.0	1.0	1.0	0.0	13.0
10	3.0	5.16	4.28	0.0	0.0	0.0	0.0	0.0	0.0	12.0

TabuLLM: AI Benchmark Report

B.3 Audit Trace: Fold 3 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.74202921834339 5	1.85391234496981 47	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	2.41909812778199 65	1.87196661429656 16	0.0	0.0	0.0	0.0	0.0	0.0	12.0
3	2.0	4.23664993041368 7	3.05497489562053 07	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	2.35802123714054 93	1.51714178698079 35	1.0	0.0	0.0	0.0	0.0	0.0	11.0
5	2.0	3.34464710997591 1	2.72092444282272 97	1.0	0.0	0.0	0.0	1.0	0.0	11.0
6	3.0	3.81701191329026 9	2.83037740827160 67	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	4.0	4.24	3.68	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	3.87	2.68	0.0	0.0	0.0	0.0	0.0	0.0	12.0
9	5.0	3.67	76.8	0.0	1.0	1.0	0.0	1.0	0.0	12.0
10	3.0	3.8	2.84	1.0	0.0	0.0	0.0	1.0	0.0	12.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.56015200960411 8	1.91209627274927 47	1.0	0.0	0.0	0.0	1.0	0.0	11.0
2	3.0	2.48998165974500 64	2.10329053940501 5	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	3.0	3.23558854319834 44	2.81925465254043 3	1.0	0.0	0.0	0.0	0.0	0.0	11.0
4	3.0	2.66950638933614 84	2.03148083199155 54	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	3.0	3.54671062475853 47	2.50664468762073 3	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	3.0	2.53606888513860 5	1.74571021622176 8	1.0	0.0	0.0	0.0	0.0	0.0	12.0
7	3.0	1.84	1.28	1.0	0.0	0.0	0.0	1.0	1.0	11.0
8	5.0	4.95	4.12	1.0	0.0	0.0	0.0	0.0	1.0	11.0
9	3.0	2.92	2.28	1.0	0.0	0.0	0.0	1.0	0.0	11.0
10	3.0	2.32	1.76	0.0	0.0	0.0	0.0	0.0	0.0	12.0

TabuLLM: AI Benchmark Report

B.5 Audit Trace: Fold 5 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.52804145398982 7	1.73286632638372 32	1.0	0.0	0.0	0.0	0.0	0.0	12.0
2	2.0	3.88209493450352 1	2.69116958683486 8	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	2.0	2.72884563893601 63	2.14515346855307 86	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	3.43568815721423 57	2.26378618407966 05	0.0	0.0	0.0	0.0	0.0	0.0	13.0
5	2.0	4.35442101400240 5	3.00297019378330 6	1.0	0.0	0.0	0.0	1.0	0.0	13.0
6	2.0	3.31999757660765 7	2.57912895967422 6	0.0	0.0	0.0	0.0	0.0	0.0	11.0
7	3.0	3.0	2.44	1.0	0.0	0.0	0.0	1.0	1.0	12.0
8	3.0	2.4	1.88	1.0	0.0	0.0	0.0	1.0	0.0	11.0
9	3.0	4.36	3.92	1.0	0.0	0.0	0.0	0.0	0.0	11.0
10	3.0	3.4	2.8	1.0	1.0	0.0	0.0	1.0	1.0	11.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	4.0	3.69838812979601 13	2.81306621323630 47	1.0	0.0	0.0	0.0	1.0	0.0	11.0
2	2.0	3.87551550715060 2	2.66299632472984 2	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	2.0	3.93856045879318 7	3.33856045879318 67	1.0	0.0	0.0	0.0	0.0	0.0	11.0
4	3.0	3.53825333377014 27	2.42707428608869 4	1.0	0.0	0.0	0.0	1.0	0.0	11.0
5	3.0	2.76211694142278 4	1.99682458786582 42	0.0	0.0	0.0	0.0	1.0	0.0	12.0
6	3.0	4.30804245507333 85	3.32575788808249 4	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	2.0	3.24	2.52	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	2.4	1.64	0.0	0.0	0.0	0.0	0.0	0.0	11.0
9	3.0	4.28	4.44	1.0	0.0	0.0	0.0	0.0	0.0	12.0
10	3.0	3.24	52.3	0.0	0.0	0.0	0.0	0.0	0.0	12.0

TabuLLM: AI Benchmark Report

B.7 Audit Trace: Fold 7 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	2.0	2.637306455352777	2.218235469659448	1.0	0.0	0.0	0.0	1.0	0.0	11.0
2	3.0	2.537741667207191 7	2.091548333441438 4	0.0	0.0	0.0	0.0	1.0	0.0	12.0
3	3.0	3.097087562282509 6	2.278058374855006 3	1.0	0.0	0.0	0.0	0.0	0.0	11.0
4	4.0	2.935655430013989 4	2.089214797194945 7	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	2.0	3.388395545652463 7	2.779692204891812	1.0	0.0	0.0	0.0	1.0	0.0	11.0
6	3.0	2.548423870741795 7	1.959741773798142 4	1.0	0.0	0.0	0.0	1.0	0.0	11.0
7	3.0	2.84	2.16	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	2.88	2.48	0.0	0.0	0.0	0.0	0.0	0.0	11.0
9	3.0	2.2	1.44	1.0	0.0	0.0	0.0	1.0	0.0	12.0
10	4.0	5.16	4.96	1.0	0.0	0.0	0.0	0.0	0.0	11.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	3.093325103634812 4	1.986502072696243 5	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	2.734268862108091	2.081403293162136 2	1.0	0.0	0.0	0.0	0.0	0.0	12.0
3	2.0	3.304755080096442	10.260515944708 97	0.0	0.0	0.0	0.0	0.0	0.0	11.0
4	4.0	3.963416272599427 5	3.137969063512 78	1.0	0.0	0.0	0.0	1.0	0.0	11.0
5	2.0	3.14327160582965	2.507102880777287	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	3.0	2.662507422739071	2.052477731782788	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	2.76	2.28	0.0	0.0	0.0	0.0	0.0	0.0	11.0
8	4.0	5.03	79.3	1.0	0.0	0.0	1.0	0.0	0.0	11.0
9	2.0	2.16	1.56	1.0	0.0	0.0	0.0	1.0	0.0	11.0
10	3.0	2.0	1.0	1.0	0.0	1.0	0.0	1.0	1.0	11.0

TabuLLM: AI Benchmark Report

B.9 Audit Trace: Fold 9 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.8871788773368547	2.191573847731785	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	2.708798520516547	2.232968581004027	0.0	0.0	0.0	0.0	0.0	0.0	11.0
3	2.0	3.2784017926954747	2.5872031372170805	1.0	0.0	0.0	0.0	1.0	0.0	11.0
4	3.0	2.9264976707230397	3.2708432450968155	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	3.0	2.7314263893816797	2.0933587905470701	1.0	0.0	1.0	0.0	1.0	0.0	11.0
6	3.0	2.44	1.135683488133095	2.0	0.0	0.0	0.0	1.0	1.0	11.0
7	3.0	3.44	3.13	1.0	0.0	0.0	0.0	1.0	1.0	12.0
8	3.0	3.64	3.12	1.0	0.0	0.0	0.0	1.0	0.0	12.0
9	3.0	2.56	60.9	0.0	0.0	0.0	0.0	0.0	0.0	11.0
10	3.0	2.8	3.36	1.0	0.0	0.0	0.0	1.0	0.0	12.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	2.0	3.9241089292555147	2.993514882092524	0.0	0.0	0.0	0.0	0.0	0.0	11.0
2	2.0	3.989385839904003	3.0952336264426745	1.0	0.0	0.0	0.0	1.0	0.0	11.0
3	3.0	2.122495588547109	1.5794938239659528	1.0	0.0	0.0	0.0	1.0	0.0	13.0
4	4.0	2.575945280782935	2.034684906405688	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	3.0	3.804050812178893	2.9620254060894466	1.0	0.0	0.0	0.0	1.0	0.0	11.0
6	3.0	2.355676091027069	1.790455199657254	1.0	0.0	1.0	0.0	1.0	0.0	11.0
7	3.0	3.48	2.52	1.0	0.0	0.0	0.0	1.0	0.0	14.0
8	3.0	4.48	3.48	0.0	0.0	0.0	0.0	0.0	0.0	12.0
9	3.0	2.63	67.3	1.0	0.0	0.0	1.0	1.0	0.0	11.0
10	4.0	5.16	4.96	1.0	0.0	0.0	0.0	0.0	0.0	11.0